

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (Electrical Engineering) (Backlog)

November 2012

EE 308 High Voltage Engineering

Date: 07.11.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain the basic circuit for high impulse current generation. [07]

Q - 2 (a) Discuss and compare half wave and full wave rectifier circuit for generation of HVDC. [08]

(b) Define the following Terms: [06]

[1] Impulse voltage [2] Wave front time [3] Wave tail time [4] Chopped impulse wave [5] Breakdown Voltage [6] Flashover voltage

OR

Q - 2 (a) A 12 - stage impulse generator has $0.126 \mu\text{F}$ capacitors. The wave front and the wave tail resistances connected are 800 ohms and 5000 ohms respectively. If the load capacitor is 1000 pF, find the front and tail times of the impulse wave produced. [07]

(b) A Cockcroft-Walton type voltage multiplier has eight stages with capacitances, all equal to $0.05 \mu\text{F}$, the supply transformer secondary voltage is 125 kV at a frequency of 150 Hz. If the load current to be supplied is 5 mA, find (a) the percentage ripple, (b) the regulation, and (c) the optimum number of stages for minimum regulation or voltage drop. [07]

Q - 3 Explain any two: [14]

[1] Tesla Coil

[2] Rogowski coil for high impulse current measurement

[3] Generating Voltmeter

SECTION - II

Q - 4 (a) Define Townsend's first and second ionization coefficients. How is the condition for breakdown obtained in Townsend's discharge? [07]

Q - 5 (a) Derive the criteria for the breakdown in electronegative gases. [08]

(b) In an experiment, in a certain gas it was found that the steady state current is 5.5×10^{-8} A at 8 kV at a distance of 0.4 cm between the plane electrodes. Keeping the field constant and reducing the distance to 0.1 cm results in a current of 5.5×10^{-9} A. calculate Townsend's primary ionization coefficient. [06]

[P.T.O.]

OR

- Q - 5 (a) What is the "stressed oil volume theory" and how does it explain breakdown in large volumes of commercial liquid dielectrics? [06]
- (b) Enlist the different mechanisms by which breakdown occurs in solid dielectrics and explain any two in detail. [08]
- Q - 6 Attempt any two: [14]

- [1] Power Frequency withstand voltage test
- [2] Impulse testing of Transformers
- [3] High voltage Schering bridge
